Savar Verma

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Education

Indian Institute of Technology, Madras — Chennai, Tamil Nadu
Bachelor of Technology in Electrical Engineering — *Batch of 2026*

Skills

Languages: JavaScript, Python, C++, C, Bash, SQL
Frameworks & Tools: Node.js, React, Vite, FastAPI, Docker, Git, Linux
Cloud & Databases: AWS (EC2), DynamoDB, PostgreSQL, Redis
Architecture & Security: OAuth 2.1, PKCE, JWT, SSO, SRP, RBAC, REST APIs, Multithreading

Work Experience

Atomberg Technologies — Feb 2026 – October 2026
Cloud & Backend Engineering Intern — *Pune, India (Hybrid)*

- Architected an **Identity Provider SaaS** on **EC2** using **Node.js** and **DynamoDB** to replace **Cognito**
- Implemented **OAuth 2.1** compliant flows with authCode, **PKCE**, and **JWT**s signed using **RS256** algorithm
- Engineered **multi-tenancy**, isolated **userpools**, and zero-knowledge **Secure Remote Password (SRP)** logins
- Integrated **Social SSO** (Google, Microsoft, and Apple) with dynamic configuration and provider toggling
- Developed a **React/Vite** admin and superadmin portal enforcing strict **Role-Based Access Model (RBAC)**
- Projected **97%** reduction in annual AWS expenditure, decreasing costs from **20 Lakh** to **50 Thousand** rupees

Projects

High-Performance Scalable URL Shortener – *Python, React, PostgreSQL* – [repo] [live] — Nov 2025

- Achieved **sub-50ms latency** by architecting a **FastAPI** system with **Redis** cache-aside strategies
- Designed a collision-free **Base62** algorithm on **Serverless PostgreSQL** to handle millions of unique records
- Engineered **asynchronous** background tasks to track real-time geo-analytics without blocking user requests
- Built a responsive **React** and **Tailwind** dashboard to visualize live system telemetry and traffic metrics

Phonetic Memory Tool – *Python, SQLite, OpenAI, Prompt Engineering* – [repo] — Sep 2026

- Built a **phonetic memory layer** that learns a user's proper-name spellings from corrected **ASR** transcripts
- Engineered **Double Metaphone** retrieval with squashed n-grams and **in-SQL TF-IDF** ranking for **5ms** lookups
- Reserved the **LLM** as a final-check gate – temperature-0, strict **JSON** after deterministic local checks
- Benchmarked **500 cases**: **87.5%** intervention accuracy with **3.2%** false positives across **157 negatives**

Multi-Threaded CLI Image Editor – *C++* – [repo] — Jul 2025

- Designed and developed a command-line **image editor** for Linux in C++ for low–level **pixel manipulation**
- Leveraged C++ multithreading to achieve a **64% performance boost** over sequential execution
- Implemented strict in-memory processing to eliminate **I/O race conditions** during concurrent batch operations

Open Source Contributions

Suckless.org (dwm) Upstream Contributor – *C, X11* – [patch 1] [patch 2] [commit] — Jan 2026 – Present

- Contributed 2 upstream patches to **dwm** (`nocornermonocle`, `fullscreen`) to optimize **X11** window logic
- Integrated **EWMH compliance** via X11 Atoms to control external compositor rendering for specific layouts
- Engineered a **state-preserving** fullscreen toggle caching layout history to resolve view switching loops

Relevant Coursework

Operating Systems, Probability and Detection Theory, Linear Algebra, Machine Learning Techniques